Curriculum Vitae – Jinho Chang

Personal Information

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Ph.D. student at KAIST, currently a member of the BioImaging-Signal Processing & Learning Lab(BISPL). I am working on diffusion and flow-matching generative models for their variational applications and inference-time techniques, and multi-modal models for informative representations and cross-modal generation. Also, leveraging my chemistry background, I aim to apply state-of-the-art AI techniques to various important chemical challenges with minimal adaptation costs.

Research Interests

-Inference-Time Scaling : Exploiting analytic and variational analyses on diffusion and flow-matching models towards training-free control of the sampling process, including reward-guided editing, negative guidance, score distillation, molecule optimization, etc.

-Alignment of Generative Models : Since training image/video generative models for specific desirable features from scratch is costly and often limited, we establish a set of reward-alignment frameworks that exploit various aspects(e.g., initial noise, intermediate guidance, stochasticity, etc.) of the base models towards task-specific external objectives while preserving prior knowledge.

-AI for Chemistry : Assessing molecular tasks that are hard to solve in principle, including conditional molecule design and editing, synthesizability and retro-synthesis prediction, intrinsic property prediction, etc.

-Multimodal Learning : Leveraging various multimodal learning of a shared data space or representation alignment, for better cross-modal understanding and generation on medical imaging and molecular structure and its properties.

Selective Publications

The full list with more details can be found in Google Scholar Page.

— **Manifold-Constrained Initial Noise Optimization for Efficient Generative Model Alignment** [link] Alignment Inference-Time Scaling
Jinho Chang, *Jong Chul Ye* *In Review, 2026*
We address existing algorithms' major challenge in black-box scenarios and introduce ZeNOVA through annealed soft-value guidance, manifold-constrained hyperspherical Langevin dynamics, and Metropolis-Hastings jumping. Experiments on image and video generative models show that ZeNOVA optimizes the initial noise toward higher rewards substantially more stably while exploiting the geometry of the Gaussian prior.

— **Reward Score Matching : Unifying Reward-based Fine-tuning for Flow and Diffusion Models** [link] Alignment
Jeongjae Lee, **Jinho Chang***, Jeongsol Kim, Jong Chul Ye* *ICML SPIGM Workshop (Best Paper), 2026*
NeurIPS, 2026
We unify reward-based fine-tuning methods for diffusion and flow models under reward score matching (RSM), which casts alignment as value-guided score matching. This view isolates key design choices - value guidance estimation and timestep-wise optimization strength - clarifying tradeoffs and enabling simpler, more efficient variants across differentiable and black-box tasks.

— **Contrastive CFG : Improving CFG in Diffusion Models by Contrasting Positive and Negative Concepts** [link] Alignment Inference-Time Scaling
Jinho Chang, *Changsun Lee, Hyungjin Chung, Jong Chul Ye* *ICML, 2026*
We point out the drawback of naively negating the Classifier-Free Guidance (CFG) term, and propose a novel approach to enhance negative CFG in conditional diffusion models using contrastive loss. Our method overcomes the distortions caused by traditional negative CFG, achieving superior sample quality and effective removal of unwanted features across various scenarios.

— **Training-Free Reward-Guided Image Editing via Trajectory Optimal Control** [link]
Alignment Inference-Time Scaling
Jinho Chang*, *Jaemin Kim**, *Jong Chul Ye* *ICLR, 2026*
Prior reward-guided sampling in diffusion and flow-matching has been studied primarily for generation and remains less explored for image editing. We introduce a training-free editing framework that formulates reward-guided editing as a trajectory optimal control problem to drive the endpoint toward higher reward. Our method outperforms inversion-based training-free baselines, achieving higher rewards while preserving source fidelity without reward hacking.

— **LDMol : Text-Conditioned Molecule Diffusion Model Leveraging Chemically Informative Latent Space** [link] AI4Chem Multimodal
Jinho Chang*, Jong Chul Ye* *ICML, 2025*
We present a new latent diffusion model LDMol for molecule generation with complex conditions like natural texts. Leveraging a chemically informative latent space obtained with contrastive learning, LDMol not only outperforms previous text-to-molecule generative models, but can also be applied to downstream tasks like retrieval and molecule editing.

— **DreamMotion : Space-Time Self-Similarity Score Distillation for Zero-Shot Video Editing** [link] Inference-Time Scaling
Hyeonho Jeong, ***Jinho Chang****, Geon Yeong Park, Jong Chul Ye* *ECCV, 2024*
We present a score distillation sampling for video editing to circumvent the standard reverse diffusion process. We propose to match the space-time self-similarities during the score distillation, demonstrating its superiority in altering appearances while accurately preserving the original structure and motion.

— **LLM-CXR : Instruction-Finetuned LLM for CXR Image Understanding and Generation** [link] Multimodal
*Suhyeon Lee**, *Wonjun Kim**, ***Jinho Chang****, Jong Chul Ye* *ICLR, 2024*
Taking inspiration from previous work on the transformer and VQ-GAN combination for bidirectional image and text generation, we developed a method for instruction-tuning a text-only LLM to gain vision-language capabilities for medical images. We let LLM understand tokenized visual inputs by instructing it to answer questions about image inputs and generate image outputs.

— **Bidirectional Generation of Structure and Properties Through a Single Molecular Foundation Model** [link] AI4Chem Multimodal
Jinho Chang*, Jong Chul Ye* *Nature Communications, 2024*
We introduce a multimodal molecular pre-trained model that integrates molecular structures and their biochemical properties, which enables various multimodal and unimodal tasks like conditional molecule generation, property prediction, molecule classification, and reaction prediction.

— **Drug-likeness Scoring Based on Unsupervised Learning** [link] AI4Chem
Jinho Chang*, Kyunghoon Lee, Seonghwan Seo, Woo Youn Kim* *Chemical Science, 2022*
We proposed a novel unsupervised learning model that could quantify the drug-likeness of a given molecule, by adopting a language model trained via unsupervised learning. Our model showed more consistent performance across different datasets and gave more reasonable scores.

Reviewer Participation

Nature Communications	*2024 -*
Communications Chemistry	*2024 -*
International Conference on Learning Representations	*2025, 2026, 2027*
Conference on Computer Vision and Pattern Recognition	*2025*
International Conference on Computer Vision	*2025*
International Conference on Machine Learning	*2026*
Neural Information Processing Systems	*2026*

Experiences

Research Intern @ HITS (https://hits.ai/) *January 2021 - June 2021*
I participated in the development of a scaffold-based molecular generative model via reinforcement learning for compounds that are both synthesizable and possess specific pharmaceutical properties.

Lab Manager @ BISPL (https://bispl-website.github.io/) *2026*
I served as the laboratory manager, overseeing overall laboratory resources and responsible research proposals, coordinating laboratory members, etc.

Rewards & Honors

Best Paper Award @ ICML 2026 4th SPIGM Workshop *2026*
Reward Score Matching : Unifying Reward-based Fine-tuning for Flow and Diffusion Models

National Science & Technology Scholarship *March 2019 - August 2022*
Ministry of Science and ICT, South Korea

KAIST Presidential Fellowship (KPF) *March 2018 - August 2022*
KAIST

Education

M.S. & Ph.D. Integrated Program in Artificial Intelligence *September 2022 - August 2028*
M.S. : 09.2022.∼08.2024. Ph.D. : 09.2024.∼08.2028.
Korea Advanced Institute of Science and Technology (KAIST)

B.S. in Computer Science and Chemistry (double major) *March 2018 - August 2022*
Korea Advanced Institute of Science and Technology (KAIST)